Integral orthogonal complements and the actual compact cross boundary

D.63. An integral compact correction lifts fifty-two more arithmetic invariant coordinates [Proved]

Keep the actual sources and maps of D.58–D.62:

$$X = \widehat{B}/(R_{50} \cap L_2), \qquad Y = X/\mathcal{K}, \qquad X_{62} = X/\mathcal{J}_{62}. \tag{D.63.1}$$

Let $P = U/\mathcal{R}$ be integral compact homology, with its actual perfect alternating intersection form. On each of the four small-character orbits of D.60, write its already constructed compact decomposition as $P_a = E_a \oplus M_a$. Here E_a is the sum of the four extra projective summands, of fibre rank 32 over the unramified quadratic coefficient ring, and M_a has fibre rank 16.

The restriction of the intersection form to every M_a is perfect. There are exact integral G-equivariant graph isomorphisms

$$\psi_a : E_a \xrightarrow{\sim} \widetilde{E}_a = M_a^{\perp}, \qquad P_a = \widetilde{E}_a \perp M_a. \tag{D.63.2}$$

They retain the kernel of each old compact projector. They have affine lifts inside the actual kernel $\mathcal{L} = \ker(U^- \to U_*^-)$, and every old central projector kernel used in $\mathcal{J}_{62}$ is unchanged. The proof supplies the actual graph, affine inclusion and retraction maps; it does not identify two differently marked embeddings without these maps.

Let $\widetilde{e}_{a,i}$ be the resulting affine projectors onto the four summands of $\widetilde{E}_a$, and let m_a be the compatible affine projector onto M_a. Both kill all cusp homology. With the cross-projector convention of D.60, put

$$\begin{aligned} \Pi_{63} &= e_\Delta \sum_{a=1}^{4} \sum_{i=1}^{4} (\overline{\widetilde{e}}_{a,i} \boxtimes \bar{m}_a), \\ \mathcal{C}_{63} &= \Pi_{63} Z, \qquad \mathcal{J}_{63} = (1 - \Pi_{63})\mathcal{J}_{62}. \end{aligned} \tag{D.63.3}$$

These are actual equivariant central submodules, and

$$\begin{aligned} \mathcal{C}_{63} \subset \mathcal{J}_{62}, &\qquad \mathcal{J}_{62} = \mathcal{C}_{63} \oplus \mathcal{J}_{63}, \\ \dim \mathcal{C}_{63} = 40960, &\qquad \dim(\mathcal{C}_{63})_G = 52, \\ \nu : (\mathcal{C}_{63})_G \xrightarrow{\sim} \mathcal{C}_{63}^G, &\qquad \bigwedge^2 \pi \text{ is injective on } \mathcal{C}_{63}. \end{aligned} \tag{D.63.4}$$

The exact orbit contributions are

ℓ	$\dim \mathcal{C}_{63,\ell}$	$\dim(\mathcal{C}_{63,\ell})_G$	four individual norm ranks
$(0,0,0,1)$	8192	12	$(4,4,4,0)$
$(0,0,1,1)$	12288	16	$(4,4,4,4)$
$(0,0,1,2)$	12288	16	$(4,4,4,4)$
$(0,1,0,0)$	8192	8	$(2,2,2,2)$

(D.63.5)

There is an actual continuous peripheral automorphism of

$$X_{63} = X/\mathcal{J}_{63}. \tag{D.63.6}$$

It has multiplier one, fixes the original cusp t, is identity on G, and retains the complete D.57 action with its literal compatible conjugators. Its full $\mathcal{W}_h$ action, normalized parameter and all established metabelian and CM comparisons through that map are unchanged.

Here is the exact change of representative on Y. Write $T = T_{61}$ for the integral homology action of D.62 and $e = \sum_{a,i} \widetilde{e}_{a,i}$. The new integral homology action is

$$T_{63} = (1 - e)T(1 - e) + e. \tag{D.63.7}$$

It is an automorphism and is identity on all $\widetilde{E}_a$. There is a continuous derivation $\eta : Y \to eU$, induced from $\mathcal{J}_\infty$, whose restriction on $U = \ker(\mathcal{J}_\infty \to G)$ is e, with $\eta(t) = 0$ and peripheral coboundaries. Define

$$\begin{aligned} A_0 &= -(1 - e)T|_{eU} : eU \longrightarrow \mathcal{L}, \\ \beta_Y(g) &= (A_0\eta(g))\alpha_{62,Y}(g). \end{aligned} \tag{D.63.8}$$

The coefficient in the second line is the actual element of the abelian kernel $\mathcal{L} \subset Y$. After the self-block corrections of D.62, the resulting Y-action is

$$\mathrm{Int}(b_{t,Y})\beta_Y, \qquad b_t = \prod_j \prod_{k=0}^{35} t^k u_j t^{-k}, \quad u_j \in S_{57}. \tag{D.63.9}$$

The vectors of the u_j are the transported D.62 correction vectors selected by the actual boundary. The element $b_{t,Y}$ centralizes t, has zero image in X_* and $\mathcal{W}_h$, and acts trivially on integral homology. Thus the projection to X_{62} is an actual newly constructed action; (D.63.8)–(D.63.9) specify its comparison with the chosen D.62 representative.

All source kernels are retained explicitly:

$$\begin{aligned} \ker(X_{63} \to X_{62}) &= \mathcal{C}_{63}, \\ \ker(X_{63} \to Y) &= (1 - e_\Delta)\mathcal{K} \oplus \mathcal{C}_{59} \oplus \mathcal{C}_{60} \oplus \mathcal{C}_{61} \oplus \mathcal{C}_{62} \oplus \mathcal{C}_{63}, \\ \ker(X \to X_{63}) &= \mathcal{J}_{63}, \\ \dim \mathcal{J}_{63} &= 48w - 2560 - 5308416\kappa - 81920. \end{aligned} \tag{D.63.10}$$

The specified small-character residual has dimension 284160 and coinvariant dimension 482. Its compact norm has rank six and kernel dimension 476. With

$$S_{63} = \{b \in R_{50} : \mathrm{im}_{\mathcal{T}_2}(b) \in \mathcal{J}_{63}\}, \qquad S_{63}/(R_{50} \cap L_2) = \mathcal{J}_{63}, \tag{D.63.11}$$

this is a nonzero remaining actual source kernel. Its boundary and the full profinite realization remain unresolved. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The compact markings and the projective types. All maps start from the marked arithmetic quotient $G \to G^\dagger = (\mathbb{Z}/3)^4 \rtimes Q \to G_*$ of D.60.2. The same forty unordered characters are used, with their original pair-coset representatives. On one character fibre, the actual affine module has rank 96 over $\mathbb{F}_4$, its cusp submodule has rank 48, its compact quotient has rank 48, and its small compact target has rank four.

The program `d_compact_splitting_maps.py` checks the actual compact quotient map, the small map and their square on every matrix column. For the old orbits the retained compact decomposition is exactly $[E_0, E_1, P_0]$, with ranks $(16, 16, 16)$. The last block is M_a; the first two split into the four rank-eight extra summands. The program checks these positions against both recorded actual affine embeddings and the compact target map. For the new orbits it uses the D.61 rank-32 projective embedding and its recorded rank-16 complement, and checks all stabilizer intertwiners.

The full unordered Sylow orders, real dimensions and norm ranks are

orbit size	$\lvert P_2\rvert$	$\dim E$	$\mathrm{rank}\, N_{P_2}\lvert E$	$\dim M$	$\mathrm{rank}\, N_{P_2}\lvert M$	$\dim \mathcal{R}$	$\mathrm{rank}\, N_{P_2}\lvert \mathcal{R}$
8	8	64	8	32	4	96	12
12	16	64	4	32	0	96	6

(D.63.12)

This uses the semilinear exchange element as well as the oriented stabilizer. To justify the projectivity test, choose vectors with independent norm images. They define a map from the corresponding number of regular modules. A nonzero kernel would contain a nonzero invariant vector, since every nonzero module of a finite two-group in characteristic two has one. The invariant vectors of a sum of regular modules are precisely its independent norm vectors, so that kernel is zero. Equality of dimensions makes the map an isomorphism. Restriction to a Sylow subgroup detects projectivity: induction followed by the coset trace splits the original module because the index is odd. This proves projectivity of all the cusp modules and all extra summands in (D.63.12). Induction over the actual pair cosets and odd averaging along the character subgroup and $G \to G^{\dagger}$ give their projective modules over the full actual group G.

In the new orbits the norm on the oriented Sylow D_8 of the complement is already zero, as checked in D.61. A nonzero projective summand would restrict to a nonzero free module with nonzero norm. Hence these complements have no projective direct summand.

In the old orbits both pieces are projective, but their indecomposable projective types are disjoint. The normal V_4-coinvariant top is a module for the odd complement C_3^2. The extra and cover weight supports, in the original two-generator marking, are

ℓ	extra support	cover support
$(0,0,0,1)$	$\{(0,1),(0,2),(1,0),(2,0)\}$	$\{(1,2),(2,1)\}$
$(0,1,0,0)$	$\{(0,1),(0,2),(1,1),(2,2)\}$	$\{(0,0),(1,2),(2,1)\}$.

(D.63.13)

The normal V_4 augmentation ideal is nilpotent, and its quotient algebra is the split semisimple algebra of C_3^2. Thus these are exactly the simple types of the projective tops. Their duals negate the weights: the V_4 socle is its norm, which the odd complement preserves. Each extra support is closed under negation and is disjoint from its cover support. Consequently the old E_a^* and M_a have no common projective direct summand. The displayed weights and their multiplicities are recomputed from all actual matrices in the certificate.

The perfect restricted form and the exact graph. Work first on the actual integral compact lattices. The character-orbit summands are mutually orthogonal: invariance under the odd character group pairs a character only with its inverse. Since each entire unordered orbit is retained, the intersection form on P_a is perfect.

Write its isomorphism $P_a \to P_a^*$ and inverse in blocks for $P_a = E_a \oplus M_a$. If the lower diagonal blocks are $D_a : M_a \to M_a^*$ and $D_a' : M_a^* \to M_a$, multiplication gives

$$D_a' D_a = 1_{M_a} - C_a' B_a. \tag{D.63.14}$$

The last endomorphism factors through E_a^*. Modulo two, an invertible Fitting summand of it would split off from both M_a and E_a^*: compose the two factor maps with the inverse on that summand to obtain the retraction and inclusion. For the new orbits this contradicts the absence of projective summands in M_a. For the old orbits it contradicts (D.63.13), after restriction to the oriented stabilizer. The Fitting summand is therefore zero, the factor endomorphism is nilpotent, and (D.63.14) is invertible modulo two. It follows that D_a is invertible integrally.

Let $C_a : E_a \to M_a^*$ be the lower off-diagonal block of the intersection form. The actual map in (D.63.2) is

$$\psi_a(v) = v - \iota_{M_a} D_a^{-1} C_a(v). \tag{D.63.15}$$

It has the same E_a-coordinate as v, pairs to zero with all M_a, and is an isomorphism onto $M_a^{\perp}$. In particular the new compact projector is $\psi_a e_{a,c}$, where $e_{a,c}$ is the old compact projector. Its kernel is still M_a. Each of the four individual projectors is transported by this same graph map.

Let $\theta_P : P \to P_*$ be the actual compact map. The perfect forms define the equivariant adjoint $\theta_P^{\dagger} : P_* \to P$, by

$$\langle \theta_P v, w \rangle_* = \langle v, \theta_P^{\dagger} w \rangle. \tag{D.63.16}$$

For each of the four fibres the certificate forms every coefficient equation $E_g H = H(P_*)_g$ for the actual oriented generators. Their rank is 128 on the 128-dimensional $\mathbb{F}_4$ space of maps from the rank-four target to the rank-32 extra part. Thus $\mathrm{Hom}_G(\bar{P}_*, \bar{E}_a) = 0$. An integral equivariant map reducing to zero is twice another such map, since the target lattice is free. Iteration and separatedness give $\mathrm{Hom}_G(P_*, E_a) = 0$. Consequently $e_{a,c}\theta_P^\dagger = 0$, so its image in this orbit lies in M_a. Equations (D.63.15)–(D.63.16) and perfection on P_* give the exact conclusion

$$\theta_P \widetilde{E}_a = 0. \tag{D.63.17}$$

Affine inclusions inside the retained source kernel. The integral cusp map is onto. Indeed the actual orders of the marked cusp in G and G_* are 36 and 12. A source boundary loop $gt^{36}g^{-1}$ maps to the cube of its target boundary loop. Every target cusp has a source preimage; multiplication by three is a unit over $\mathbb{Z}_2$, and the single sum relation maps to the corresponding sum relation. These literal boundary maps prove surjectivity, also after the odd character projectors. The finite certificate checks its rank four on every target cusp fibre.

The diagram with exact rows $0 \to \mathcal{R} \to U \to P \to 0$ and its small counterpart therefore gives an onto map

$$\ker \theta_U \longrightarrow \ker \theta_P. \tag{D.63.18}$$

Its restriction to the active part is the same map with source $\mathcal{L}$. Projectivity of $\widetilde{E}_a$, which is integrally isomorphic to E_a, lifts its inclusion in (D.63.17) through (D.63.18). Denote the resulting inclusion by $j_a : \widetilde{E}_a \hookrightarrow \mathcal{L}$. Compactification is exactly the prescribed inclusion, so this is a split injection of free lattices. This proves the affine small-map zero without postulating a stronger affine Hom vanishing.

Write $U_\Sigma, \mathcal{R}_\Sigma, P_\Sigma$ for the sums of these four character components. On this specified part the cusp lattice is integrally projective. Its projective reduction in (D.63.12) lifts to an idempotent projective lattice; projectivity lifts an isomorphism modulo two onto the actual cusp lattice. The resulting map is an isomorphism by its invertible reduction. The dual cusp lattice is also projective, since the dual of a regular group-ring lattice is regular. Dualizing the affine exact sequence therefore splits it, and dualizing back supplies a retraction $r : U_\Sigma \to \mathcal{R}_\Sigma$. This argument requires no projectivity of P.

Replace this retraction by

$$r' = r - \sum_a r j_a \widetilde{e}_{a,c} \pi. \tag{D.63.19}$$

It is still identity on $\mathcal{R}_\Sigma$ and vanishes on all $j_a \widetilde{E}_a$. Thus $1 - r'$ induces a compact section $j_P : P_\Sigma \to U_\Sigma$ agreeing with the chosen j_a on these summands. Set

$$\widetilde{e}_{a,i} = j_a \psi_a e_{a,i,c} \pi, \qquad m_a = j_P(1 - \widetilde{e}_{a,c})\pi \quad \text{on the } a\text{th orbit.} \tag{D.63.20}$$

Extend the projectors by zero on all other character components. These are mutually orthogonal equivariant affine idempotents; they kill cusps and inject under compactification. Their reductions have precisely the old module matrices in the transported bases.

There is an explicit comparison with the old affine marking. Let e^{old} be the sum of its sixteen old projectors, and use the old compact inclusion to interpret $\psi\pi e^{\mathrm{old}}$. Then

$$\Psi = 1 - e^{\mathrm{old}} + j\psi\pi e^{\mathrm{old}} : U \to U \tag{D.63.21}$$

is an integral equivariant automorphism, with $(\Psi - 1)^2 = 0$, $\theta_U \Psi = \theta_U$, and $\Psi|_{\mathcal{R}} = 1$. Its new projectors satisfy $\widetilde{e}_{a,i} = \Psi e^{\mathrm{old}}_{a,i}$, and the old projectors satisfy $e^{\mathrm{old}}_{a,i}\Psi = e^{\mathrm{old}}_{a,i}$. The exterior map $\bigwedge^2 \Psi$ carries every old selected self or cross image to its new image and has the same kernel on that projector. It also satisfies $\theta_Z \bigwedge^2 \bar{\Psi} = \theta_Z$, so it preserves the actual comparison kernel $\mathcal{K}$. It fixes the other character orbits. In particular the literal submodule $\mathcal{J}_{62}$ is unchanged.

The integral homology action and its actual peripheral realization. D.60–D.62 give $e^{\mathrm{old}}_{a,i}T = e^{\mathrm{old}}_{a,i}$. Hence $eT = e$ for the new total projector. The kernel $\ker e$ is invariant under T, and its restriction is an automorphism, since T is an automorphism and induces identity on the quotient eU. Relative to $U = eU \oplus \ker e$, its matrix is $\left(\begin{smallmatrix} 1 & 0 \\ C & D \end{smallmatrix}\right)$. Formula (D.63.7) replaces this by $\left(\begin{smallmatrix} 1 & 0 \\ 0 & D \end{smallmatrix}\right)$. Thus T_{63} is invertible, fixes all cusps, retains $\theta_U T_{63} = T_* \theta_U$, and its difference from T is $A_0 e$. The image of A_0 is in $\mathcal{L}$ because $\theta_U eU = 0$ and the old small square commutes.

The projective module eU has zero positive cohomology over the finite group G. Inflation–restriction therefore lifts the actual equivariant map $e : U \to eU$ to a continuous derivation of $\widehat{B}$. Its value on t^{36} is zero, since e kills cusp homology. On the projective cyclic module, the exact norm sequence gives $\ker N_{36}(t) = (1-t)eU$; subtract the corresponding inner derivation to make its value on t zero. The same projective norm sequence for s and r gives actual peripheral values

$$\eta(s) = (1-s)d_s, \qquad \eta(r) = (1-r)d_r, \qquad \eta(t) = 0, \qquad d_s, d_r \in eU. \tag{D.63.22}$$

All these assertions hold over $\mathbb{Z}_2$: the regular-module cyclic identities split, and direct summands preserve their exactness. The derivation kills the kernel of the full homology source $\mathcal{J}_\infty$ of D.50, so it factors through Y.

Composing with the actual map $A_0 : eU \to \mathcal{L}$ gives a derivation with values in the abelian kernel of $Y \to X_*$. Since $\alpha_{62,Y}$ is identity on G, the multiplication formula in (D.63.8) is a homomorphism. Its peripheral conjugators change by $A_0 d_s, A_0 d_r$, which have zero images in X_* and $\mathcal{W}_h$. It fixes t and induces the invertible matrix T_{63}. Its maps on the quotient and on the kernel in the D.58 fibre product are automorphisms, so it is an actual automorphism of Y. Each new conjugating element has an actual lift in S_{57} by D.58.2. Choosing these lifts gives the literal source conjugators used below.

The cross-module norms and the actual boundary zero. In a transported fibre, a cross block has coordinates $\mathrm{Mat}_{8\times 16}(\mathbb{F}_4)$, of dimension 256 over $\mathbb{F}_2$. An oriented stabilizer element acts by $H \mapsto E_g H \overline{M_g}^{\,t}$; a character exchange applies coefficient conjugation as well. The program `d_all_compact_cross_norm.py` forms every such column, all relations $(g-1)H$, and their complete group norm. It verifies that the norm kills every relation and that every norm column is invariant. The quotient dimension equals the norm rank in each of all sixteen blocks, with the values in (D.63.5). Induction over the actual pair cosets multiplies dimensions by eight or twelve and leaves coinvariants and norm ranks as displayed. This proves (D.63.4).

The compact action of T_{63} is block diagonal on the actual orthogonal decomposition $\widetilde{E}_a \perp M_a$, and is identity on the first part. The compact orientation bivector has zero cross projection there. Its image under $\bigwedge^2 \overline{T}_{63}$ has the same zero projection. For a preliminary source lift with $\varphi(t) = zt$, D.55's full, unquotiented orientation identity therefore gives

$$(\bigwedge\nolimits^{2} \pi)\nu([\Pi_{63} z]) = 0. \tag{D.63.23}$$

Both maps on this selected module are injective by (D.63.4), so the actual cross boundary class is zero. This evaluates the boundary through its actual orientation map, rather than transferring a conclusion from the separate order-1080 example.

The chosen affine complement can have cusp terms under T_{63}. This does not affect descent: its projected action satisfies

$$\widetilde{e}_{a,i} T_{63} = \widetilde{e}_{a,i}, \qquad m_a T_{63} = D_{a,c} m_a, \tag{D.63.24}$$

where $D_{a,c}$ is the compact complement action expressed on the chosen affine section. The second formula follows by compactification; both sides kill cusps and all other components. Thus the kernel of each cross projector is invariant, with invertible induced map on its image.

Retaining the earlier coordinates and correcting the source. The graph comparison (D.63.21) preserves every old central projector kernel. Moreover the sixteen new projectors have fixed left rows for T_{63}. The D.60 and D.61 selected self and cross norms therefore still force zero boundary on their injective-norm components. The D.59 selected character orbits are unchanged.

For the remaining twenty-six self coordinates, transport the exact D.62 maps by Ψ:

$$[v] \longmapsto [e_{\Delta}(v \wedge sv)]_G, \qquad [v] \longmapsto [e_{\Delta}(v \wedge N_{36}(t)v)]_G. \tag{D.63.25}$$

The homology vectors now lie in the actual $\widetilde{E}_{a,i} \subset \mathcal{L}$; their matrices, induced coset bases and all twenty-six independent columns are unchanged through the proved module isomorphism. D.58.2 supplies their actual preimages $u_j \in S_{57}$. The source involution words $u\widetilde{s}(u)$ and cusp words $\prod_{k=0}^{35} t^k u t^{-k}$ are exactly those of D.62, with the new source conjugators. In particular the wrap term is still $t^{36}ut^{-36}$, not u. Their errors lie in the selected self block plus terms with a cusp factor. Hence their projection to $\mathcal{C}_{63}$ is zero. They cancel the whole self norm kernel, change the Y-action by precisely (D.63.9), and retain its homology.

Every now retained central boundary class is zero. The exact augmentation map $(b, c) \mapsto (1 - s)b + s(1 - r)c$ realizes its remaining representative as actual central changes of the two conjugators. These corrections preserve Y and the complete X_* conjugators. The final cusp error lies in $\mathcal{J}_{63}$. The defining class-two laws and the projector kernel identities give descent through that submodule. The invertible Frattini map gives an automorphism of the finite class-two component; its common quotient with the full homology and $\mathcal{W}_h$ sources remains equivariant. Their actual marked fibre product is exactly $X/\mathcal{J}_{63}$. This proves (D.63.6).

The actual remaining kernel and the computation record. The image of every cross projector is in $\mathcal{J}_{62}$: its first factor has zero small image, its second is killed by every old extra projector, and its character support is disjoint from the D.59 support. Thus (D.63.3) is a direct sum decomposition inside the literal old source kernel. Subtracting its dimensions gives

$$325120 - 40960 = 284160, \qquad 534 - 52 = 482. \tag{D.63.26}$$

The original compact norm rank in D.60 was 138. D.60, D.61 and D.62 removed ranks 28, 50 and 2, respectively. The present compact-injective norm image removes rank 52, leaving

$$138 - 28 - 50 - 2 - 52 = 6, \qquad 482 - 6 = 476. \tag{D.63.27}$$

The compact projectors make these images direct: they kill the complementary residual and are identity on their own compact norm images. Consequently these are ranks of the actual remaining comparison maps, not dimension estimates. Taking a nonzero residual vector and its source preimage under $R_{50} \to Z$ gives (D.63.11) and proves that the remaining kernel is nonzero.

The complete programs also retain a separate new-orbit cross calculation and the cusp/complement module calculation, with every original output. Two abandoned old-orbit extractions are preserved in the trial record: one assumed zero cross coinvariants, and another used the wrong position of P_0 in the recorded basis. Neither is used in this proof. The final verifier checks that position against the actual target and affine maps before extracting any action. Its first corrected mathematical run had a wall-clock variable collision; that complete run is preserved, and the fresh timing-corrected run has identical mathematical fields. The separate splitting verifier's initial missing input-key error is also retained. The final four programs and all their complete outputs are replayed independently from copied immutable inputs. □